(b)	**Properties of waves**
Students should:	
3.2	explain the difference between longitudinal and transverse waves
3.3	know the definitions of amplitude, wavefront, frequency, wavelength and period of a wave
3.4	know that waves transfer energy and information without transferring matter
3.5	know and use the relationship between the speed, frequency and wavelength of a wave: wave speed = frequency × wavelength $v = f \times \lambda$
3.6	use the relationship between frequency and time period: $\text{frequency} = \frac{1}{\text{time period}}$ $f = \frac{1}{T}$
3.7	use the above relationships in different contexts, including sound waves and electromagnetic waves
3.8	explain why there is a change in the observed frequency and wavelength of a wave when its source is moving relative to an observer and that this is known as the Doppler effect
3.9	explain that all waves can be reflected and refracted